Theoretical Aspects of Modern Cryptography

Homework 2

2026 Spring

Homework rules

- Group discussion is *highly recommended*. But you have to write down your solution by *yourself*. Aim for clarify and conciseness in your solution, emphasizing the main ideas over low-level details.
- You should *cite* all the references that help you to solve the homework, including the people you have discussed with and the online material you looked at.
- You are required to type your solution using latex and submit a *pdf* to Gradescope and a *tex* file to NTUCool. You may type your solution in English (preferable) or Chinese.
- You are not expected to solve *all* problems. Just try your best! If you don't solve a problem, try your best to write down your thinking process.
- You may use AI however you wish to deepen your understanding of the lecture material. Upload the notes, talk to your AI about them, ask for more explanation or examples; it's all fine. You may not use LLMs in any way to work on your homework. You may not upload assignments, ask for hints, ask how certain concepts from the lectures might be applied to specific homework problems, or upload your assignments to check for correctness or clarity or anything else. You may not include any AI generated content whatsoever in your homework submissions (This policy is taken from MIT 6.5620).
- We will use a *random-checking policy* in which students may be asked to present their homework solutions during TA office hours, with the expectation that everyone will present *at least once*. Evaluation is based on *consistency*, meaning students must be able to reproduce the solutions they originally submitted. If a student fails this consistency check, they will receive a significant *percentage deduction* from their homework score.
- 作業總分為41分，可得最大計分為35分。
- Deadline 4/26(日)23:59，遲交時間最晚4/28(二) 23:59 (遲交成績打9折)，超過則以0分計算。

1. (6 points)**(Fun with indistinguishability)** In this problem, we will use the concrete version of computational indistinguishability. We say that two distribution X, Y is (t, ϵ)-indistinguishable, denoted as $X \approx_{t,\epsilon} Y$, if for any distinguisher D that runs in time t,

$$|\Pr[\mathcal{D}(x) = 1 : x \leftarrow X] - \Pr[\mathcal{D}(y) = 1 : y \leftarrow Y]| < \epsilon.$$

Here, the probability notation is interpreted as follows: the expression to the right of the colon describes the experiment (i.e., the sampling process for the random variable), and the expression to the left specifies the event. In other words, $\Pr[\text{Event} : \text{Experiment}]$ denotes the probability that the event occurs when the experiment is executed.

While it is not requested, you are also encouraged to think about the asymptotic version of the following problems.

In this problem, you are allowed to use the computational assumptions taught in class. For instance, you are allowed to assume the existence of (t, ϵ)-OWF for some (t, ϵ).

   (a) (2 points) Let $\mathcal{A}$ be some algorithm that runs in time T. Show that if $X \approx_{t,\epsilon} Y$ for some $t > T$, then $\mathcal{A}(X) \approx_{(t-T),\epsilon} \mathcal{A}(Y)$.

   (b) (2 points) Following the previous question, show that $\mathcal{A}(X) \approx_{t,\epsilon} \mathcal{A}(Y)$ is not necessarily true.

   (c) (2 points) Let X, Y, Z be three arbitrary distributions. Show that $X \approx_{t,\epsilon} Y$ does not necessarily imply $(X, Z) \approx_{t,\epsilon} (Y, Z)$

2. (9 points) **(Fun with PRF)** Assuming the existence of PRFs. Prove the statement "for all pseudorandom functions $F'_{k'} : \{0,1\}^n \to \{0,1\}^n$ with $k' \in \{0,1\}^n$ as the key, F_k is a pseudorandom function", or provide a counterexample to demonstrate that it is not true. As before, $\oplus$ denotes bitwise XOR, and $x_{[a:b]}$ denotes the substring of x from the a-th bit to the b-th bit.

(a) (3 points) $F_k(x) = F'_{k_1}(F'_{k_2}(x))$, where $k = (k_1, k_2)$. (Hint: Hybrid argument)

(b) (3 points) $F_k(x) = F'_k(x) \oplus x$.

(c) (3 points) $F_k(x) = F'_k(x) \oplus k$.

3. (6 points) **(Everything is One-Way function)** In the lecture, we demonstrated (or claimed) that OWF implies PRG, and PRG implies PRF. While this shows the possibility of building cryptographic primitives from OWF, it does not fully explains why we start from OWFs. What if, for instance, OWF is in fact a much stronger primitive comparing to PRG?

   In this problem, you will be asked to fill up the gap by showing that the inverse direction of these statements are both **easy** and **direct**. In other words, prove the following statements with simple constructions:

   (For simplicity, in problem (a) and (c) you can assume the existence of a PRF $f : \{0,1\}^{l(n)} \times \{0,1\}^n \to \{0,1\}^n$, where $l(n) = \Theta(n^c)$ for some constant $c > 0$)

   (a) (2 points) PRF implies OWF ;

   (b) (2 points) PRG implies OWF ;

   (c) (2 points) PRF implies PRG. (You are not supposed to prove this through claiming PRF $\implies$ OWF $\implies$ PRG)

4. (6 points) **(Extending GGM: Larger Range and Flexible Inputs)** Let us first recall the GGM construction. Given a PRG $G : \{0,1\}^n \to \{0,1\}^{2n}$, where we define $G_0, G_1 : \{0,1\}^n \to \{0,1\}^n$ such that $G(\cdot) = G_0(\cdot)\|G_1(\cdot)$, we learned that we can construct an PRF $F : \{0,1\}^n \times \{0,1\}^n \to \{0,1\}^n$ where

$$F_k(x) = G_{x_n} \circ G_{x_{n-1}} \circ \cdots \circ G_{x_1}(k)$$

   In the GGM construction described above, the PRF F_k maps n-bit inputs to n-bit outputs. However, in general, there is no requirement that a PRF must restrict both its input and output lengths to n bits. It is straightforward to extend the above construction to support inputs of length $t(n)$, for some polynomial $t(\cdot)$, as follows:

$$F_k(x) = G_{x_{t(n)}} \circ G_{x_{t(n)-1}} \circ \cdots \circ G_{x_1}(k). \tag{1}$$

   The security of the above construction is identical to that of the original case. However, even in this extension, F_k only supports inputs of a fixed length $x \in \{0,1\}^{t(n)}$, and the output length remains n bits.

   In the following problem, you are asked to construct PRFs that support multiple input lengths, where each input length may correspond to a different output length. Note that the security definition of a PRF with varying input and output lengths can be naturally extended by considering a random function with appropriately defined input and output domains.

   (a) (2 points) The first attempt is to use the same construction to directly support multiple input lengths:

$$F_k(x) = G_{x_{|x|}} \circ G_{x_{|x|-1}} \circ \cdots \circ G_{x_1}(k).$$

   Show that the above construction is not secure, even when it supports only two input lengths, namely $\{0,1\}^n$ and $\{0,1\}^{n-1}$.

   (b) (4 points) We say PRF F_k support (a, b) query if for any $x \in \{0,1\}^a$, then $F_k(x) \in \{0,1\}^b$. Construct a PRF that can support an arbitrary $\ell(n)$ distinct (a_i, b_i) queries, where l is some polynomial and for all $i \in [\ell(n)]$, both a_i and b_i are also bounded by some polynomial in n. (You may assume the existence of a pseudorandom generator (PRG) with arbitrary polynomial stretch and construction in Equation (1) is a secure PRF.)

5. (3 points) **(Complete the proof in the class)** Let $G : \{0,1\}^n \to \{0,1\}^{2n}$ be a PRG. Recall that in order to show that

$$(G(U_n^{(1)}), \ldots, G(U_n^{(t)})) \simeq_c (U_{2n}^{(1)}, \ldots, U_{2n}^{(t)})$$

for $t = \mathsf{poly}(n)$, where $\{U_n^{(1)}, \ldots, U_n^{(t)}\}$ (resp. $\{U_{2n}^{(1)}, \ldots, U_{2n}^{(t)}\}$) is an independent collection of copies of U_n (resp. U_{2n}), we first constructed the distribution

$$\mathsf{Hyb}_i = (U_{2n}^{(1)}, \ldots, U_{2n}^{(i)}, G(U_n^{(i+1)}), \ldots, G(U_n^{(t)}))$$

for each $i \in [t]$ as hybrids. Suppose to the contrary that there exists a PPT distinguisher $\mathcal{D}$ such that for infinitely many n

$$|\Pr[\mathcal{D}((G(U_n^{(1)}, \ldots, G(U_n^{(t)})) = 1] - \Pr[\mathcal{D}((U_{2n}^{(1)}, \ldots, U_{2n}^{(t)})) = 1] > \frac{1}{n^c}$$

for some c. By Hybrid Lemma there exists an i such that for infinitely many n

$$|\Pr[\mathcal{D}(\mathsf{Hyb}_i) = 1] - \Pr[\mathcal{D}(\mathsf{Hyb}_{i+1}) = 1] > \frac{1}{tn^c}.$$

In the class we then constructed a reduction R, supposing that R knows such an i, to break the pseudo-randomness of G, i.e., we have showed that for such an R it holds that

$$\Pr[R \textit{ wins}] > \frac{1}{2} + \frac{1}{2tn^c}.$$

In fact, there is no reason to assume that R knows i. However one can modify the reduction R' as follows:

(i) when receiving a challenge x from the challenger $\mathbf{Ch}$ in the PRG game, sample $i \xleftarrow{\$} [t]$,

(ii) sample $k_1, \ldots, k_i \xleftarrow{\$} U_{2n}$, $k_{i+2}, \ldots, k_t \xleftarrow{\$} U_n$

(iii) send $(k_1, \ldots, k_i, x, G(k_{i+2}), \ldots, G(k_t))$ to the distinguisher $\mathcal{D}$.

(iv) when receiving the answer b' from $\mathcal{D}$, output b' back to $\mathbf{Ch}$.

Show that such a reduction R' can also break the pseudo-randomness of G with the same advantage as R, i.e.,

$$\Pr[R' \textit{ wins}] > \frac{1}{2} + \frac{1}{2tn^c}.$$

6. (11 points) **(Commitment Scheme from PRG)** Informally, *commitment schemes* are usually referred to as a digital equivalent of a "locked" box. They consist of two phases:

- **Commit Phase**: Sender puts a value v in a locked box;
- **Reveal Phase**: Sender unlocks the box and reveal v.

Intuitively, we require that before the reveal phase the value v should remain hidden: this property is called *hiding*. Additionally, during the reveal phase, there should only exists a single value that the commitment can be revealed to: this is called *binding*. In general, the commit phase can be interactive (i.e., the sender and receiver can exchange messages). The reveal phase is always non-interactive (i.e., the sender sends one message to the receiver). If the commit phase is interactive (non-interactive), we call the interactive (non-interactive) commit.

In this problem, we will present an interactive commitment scheme from PRG and ask you to prove its hiding and binding properties. Before we present the construction, we will need to define some basic notation, including the syntax of the interactive protocol and the security of the interactive commitment scheme. We first define what an interactive protocol is and what the view is.

Definition 1 (Two-party interactive protocol) A two-party interactive protocol $\langle S, R\rangle$ is a pair of PPT algorithms. For inputs x, y and random coins r_S, r_R, we denote an execution by:

$$\langle S(x; r_S), R(y; r_R)\rangle \;\to\; (\mathsf{out}_S, \mathsf{out}_R, \tau),$$

where τ is the transcript of exchanged messages, out_S and out_R are the outputs of S and R, respectively.

Remark 1 The above definition uses a compact notation that hides the round-by-round interaction between S and R. Concretely, the protocol proceeds in multiple rounds, where each message is computed as a function of the party's input, its random coins, and the transcript so far. One can equivalently define a protocol by explicitly specifying the message-generation functions. In this problem, we adopt the compact notation for simplicity.

Definition 2 (View of both parties) Let τ be the transcript generated in the execution above. The views of the parties are defined as:

$$\mathsf{view}_S^{\langle S,R\rangle}(x, y; r_S, r_R) := (x, r_S, \tau) \text{ for sender's view},$$

$$\mathsf{view}_R^{\langle S,R\rangle}(x, y; r_S, r_R) := (y, r_R, \tau) \text{ for receiver's view}.$$

When r_S, r_R are sampled uniformly, we write $\mathsf{view}_S^{\langle S,R\rangle}(x, y)$ and $\mathsf{view}_R^{\langle S,R\rangle}(x, y)$ for the induced random variables.

Remark 2 The notion of a view formalizes all the information available to a party at the end of an execution of the protocol. From the definition of $\mathsf{view}_R^{\langle S,R\rangle}$ (receiver's view), we see that it consists of the receiver's input y, its internal randomness r_R, and the transcript τ of exchanged messages. This captures all the information obtained by R during the execution. Therefore, no additional internal state of R needs to be included in the definition of $\mathsf{view}_R^{\langle S,R\rangle}$.

Given the definition of view and interactive protocol, we can define the syntax of the interactive commitment scheme.

Definition 3 (Syntax of interactive commitment) An interactive commitment $(\langle S, R\rangle, \mathsf{Open})$ consists of the two-phase commit phase and reveal phase. We first define the syntax of $(\langle S, R\rangle, \mathsf{Open})$.

- $\langle S(b, 1^n), R(1^n)\rangle \to (d_S, \tau)$: This is an interactive protocol between a PPT sender S with input bit $b \in \{0, 1\}$ and security parameter 1^n, and a PPT receiver R with input security parameter 1^n. At the end of the execution:
  - the sender S outputs d_S, which is the decommitment (opening) information for b,
  - τ denotes the transcript of the execution.
- $\mathsf{Open}(b, d_S, \tau) \to \{\top, \bot\}$: This is a deterministic algorithm that takes as input the sender's opening information d_S and the opening bit b, along with the transcript τ. The algorithm outputs $\top$ if the opening is valid. Otherwise, the algorithm will output $\bot$ to indicate the opening has failed.

We now define the commit phase and the reveal phase in terms of $(\langle S, R\rangle, \mathsf{Open})$.

Commit Phase: The sender on input $(b, 1^n)$ and the receiver on input 1^n execute the interactive protocol $\langle S(b, 1^n), R(1^n)\rangle$, resulting in output (d_S, τ).

Reveal Phase: The sender sends (b, d_S) to the receiver. The receiver computes $\mathsf{Open}(b, d_S, \tau)$ and accepts the opening if and only if the output is $\top$.

We require the correctness property for $(\langle S, R\rangle, \mathsf{Open})$.

Definition 4 (correctness) An interactive commitment scheme $(\langle S, R\rangle, \mathsf{Open})$ is correct if $\forall b \in \{0, 1\}$ and $\forall n \in \mathbb{N}$, the following hold

$$\Pr\left[\mathsf{Open}(b, d_S, \tau) = \top : (d_S, \tau) \leftarrow \langle S(b, 1^n), R(1^n)\rangle\right] = 1,$$

where the probability is taken over both the sender's and receiver's randomness. Here, the probability notation is interpreted as follows: the expression to the right of the colon describes the experiment (i.e., the sampling process for the random variable), and the expression to the left specifies the event. In other words, $\Pr[\text{Event} : \text{Experiment}]$ denotes the probability that the event occurs when the experiment is executed.

Next, we describe the hiding and binding security of the interactive commitment scheme.

Definition 5 (hiding) An interactive commitment scheme is *computationally hiding* (resp., *statistically hiding*) if for every PPT (resp., unbounded) receiver R^* and every PPT (resp., unbounded) distinguisher D, for all $n \in \mathbb{N}$, the following holds:

$$\left|\Pr\left[D(\mathsf{view}_{R^*}^{\langle S(0,1^n),R^*(1^n)\rangle}) = 1\right] - \Pr\left[D(\mathsf{view}_{R^*}^{\langle S(1,1^n),R^*(1^n)\rangle}) = 1\right]\right| \leq \mathsf{negl}(n),$$

where $S(b, 1^n)$ denotes the honest sender with input b and security parameter 1^n.

Remark 3 The hiding property captures security against a malicious receiver. In this setting, the sender S is honest, while the receiver R^* may arbitrarily deviate from the protocol.

Remark 4 Intuitively, the goal of the malicious receiver is to distinguish whether the sender committed to $b = 0$ or $b = 1$ based on its view of the interaction. The scheme is computationally hiding if, for every PPT adversarial receiver R^*, the views corresponding to $b = 0$ and $b = 1$ are computationally indistinguishable.

Definition 6 (binding) The interactive commitment scheme is *computationally binding* (resp., *statistically binding*) if for every PPT (resp., unbounded) sender S^*, for all $n \in \mathbb{N}$, the following holds:

$$\Pr\left[\begin{array}{c}\mathsf{out}_{S^*} = (d_{S^*}^0, d_{S^*}^1) \\ \mathsf{Open}(0, d_{S^*}^0, \tau) = \mathsf{Open}(1, d_{S^*}^1, \tau) = \top\end{array} : (\mathsf{out}_{S^*}, \mathsf{out}_R, \tau) \leftarrow \langle S^*(1^n), R(1^n)\rangle\right] \leq \mathsf{negl}(n),$$

where $R(1^n)$ denotes an honest receiver with input the security parameter 1^n.

Remark 5 The binding property captures security against a malicious sender. In this setting, the receiver R is honest, while the sender S^* may arbitrarily deviate from the protocol.

Remark 6 Intuitively, the goal of a malicious sender S^* is to produce two valid openings $d_{S^*}^0$ and $d_{S^*}^1$, one corresponding to $b = 0$ and the other to $b = 1$. The scheme is computationally binding if, for every ppt sender S^*, they cannot find two valid openings $d_{S^*}^0$ and $d_{S^*}^1$.

Finally, we can describe our construction of an interactive commitment scheme from PRG. Let $G : \{0, 1\}^n \to \{0, 1\}^{3n}$ be a PRG. The construction of interactive commitment scheme $(\langle S, R\rangle, \mathsf{Open})$ is defined as follows:

Construction of $\langle S(b, 1^n), R(1^n)\rangle \to (d_S, \tau)$ **:**

(1) R sample a random string $r \in \{0,1\}^{3n}$ and send r to S.

(2) S receives the message r. Then S samples a random string $s \in \{0,1\}^n$ and compute $c \leftarrow \mathsf{Com}(s, r, b)$ by

$$c = \mathsf{Com}(s, r, b) := \begin{cases} G(s) & \textit{if } b = 0 \\ G(s) \oplus r & \textit{if } b = 1 \end{cases}.$$

Lastly, S sends c back to R and output s as its opening information d_S.

(3) The transcript $\tau := (r, c)$ which include the first message r and the second message c.

Construction of $\mathsf{Open}(b, d_S, \tau) \to (\top, \bot)$**:**

(1) Parse $\tau := (r, c)$.

(2) If $\mathsf{Com}(d_S, r, b) = c$, then output $\top$ and otherwise output $\bot$.

Below are the problem statements.

(a) (2 points) Show that the above scheme satisfies the correctness.

(b) (2 points) Show that the above scheme is computational hiding.

(c) (3 points) Show that the above scheme is statistically binding.

(d) (4 points) Show that there is no interactive commitment scheme that satisfies both statistical hiding and statistical binding.